Final Analyzed Data: Project2-Data.csv (cleaned & merged dataset)
Raw Data Source: Project2-RawData.csv
Unit of Observation: Each row represents a monthly observation (Year–Month) spanning 1980–2025.

554 observations

1. Year (Month-Year)

- **Source & Processing:**
  Provided in the raw dataset and used as the key to merge CPI and Consumer Sentiment data. Converted into a consistent datetime format.
- **Observations: ~554**
- **Data Type: Time index**
- **Use in Analysis:**
  Enables time-series modeling and alignment across datasets

2. Consumer Sentiment Index (UMCSENT)

- **Source & Processing:**
  Collected from FRED (University of Michigan Survey of Consumers). Merged with CPI data by date.
- **Observations: ~554**
- **Data Type: Quantitative (continuous)**
- **Role: Dependent variable in all models (OLS, Lasso, ARIMAX)**
- **Description: Measures consumer optimism/pessimism about the economy**

3. Overall CPI

- **Source & Processing:**
  Collected from the U.S. Bureau of Labor Statistics (BLS) as part of the CPI dataset. Merged with Consumer Sentiment data using monthly timestamps. Standardized during preprocessing for consistency with other CPI components.
- **Observations: ~554 (aligned with full dataset)**
- **Data Type: Quantitative (continuous)**
- **Role: Used as a baseline measure of overall inflation. Provides a benchmark to compare against individual CPI components (e.g., Food, Shelter)**

3. Food

- **Source & Processing:**
  Obtained from BLS CPI dataset. Standardized during preprocessing.
- **Observations: ~554**
- **Data Type: Quantitative**
- **Role: Independent variable**
- **Notes:**
  Identified as the most significant negative predictor of consumer sentiment across all models

4. Shelter

- **Source & Processing:**
  BLS CPI dataset. Standardized and retained after feature selection.
- **Observations: ~554**
- **Data Type: Quantitative**
- **Role: Independent variable**
- **Notes:**
  Initially hypothesized as primary driver but found less impactful than food

5. Apparel

- **Source & Processing:**
  Obtained from BLS CPI dataset. Standardized during preprocessing.
- **Observations: ~554**
- **Data Type: Quantitative**
- **Role: Independent variable**
- **Notes:**
  - **Represents Discretionary consumer spending**

6. Core CPI

- **Source & Processing:**
  Obtained from BLS CPI dataset. Standardized during preprocessing.
- **Observations: ~554**
- **Data Type: Quantitative**
- **Role: Benchmark inflation measure**

- **Notes:**
    - **Used to capture underlying inflation trends**

7. Gasoline (all types)

- **Source & Processing:**
  Obtained from BLS CPI dataset. Standardized during preprocessing.
- **Observations: ~554**
- **Data Type: Quantitative**
- **Role: Independent variable**
- **Notes:**
    - **Highly volatile**
    - **Strong psychological impact due to visibility of price changes**

8. Electricity

- **Source & Processing:**
  Obtained from BLS CPI dataset. Standardized during preprocessing.
- **Observations: ~554**
- **Data Type: Quantitative**
- **Role: Independent variable**
- **Notes:**
    - **Represents essential household costs**
    - **Moderately stable compared to gasoline**

9. Natural Gas

- **Source & Processing:**
  Obtained from BLS CPI dataset. Standardized during preprocessing.
- **Observations: ~554**
- **Data Type: Quantitative**
- **Role: Independent variable**
- **Notes:**
    - **Seasonal variation**

10. Commodities: Less Food and Energy Commodities

- **Source & Processing:**
  Obtained from BLS CPI dataset. Standardized during preprocessing.
- **Observations: ~554**
- **Data Type: Quantitative**
- **Role: Independent variable**

- **Notes:**
    - **Captures non-essential goods pricing**

11. New Vehicles

- **Source & Processing:**
  Obtained from BLS CPI dataset. Standardized during preprocessing.
- **Observations: ~554**
- **Data Type: Quantitative**
- **Role: Independent variable**
- **Notes:**
    - **Durable goods category**
    - **Sensitive to suppl chain disruptions**

12. Medical Care Commodities

- **Source & Processing:**
  Obtained from BLS CPI dataset. Standardized during preprocessing.
- **Observations: ~554**
- **Data Type: Quantitative**
- **Role: Independent variable**
- **Notes:**
    - **Represents healthcare goods (e.g., prescriptions)**
    - **Less volatile than energy categories**

13. Services Less Energy Services

- **Source & Processing:**
  Obtained from BLS CPI dataset. Standardized during preprocessing.
- **Observations: ~554**
- **Data Type: Quantitative**
- **Role: Independent variable**
- **Notes:**
    - **Broad services category excluding energy**

14. Medical Care Services

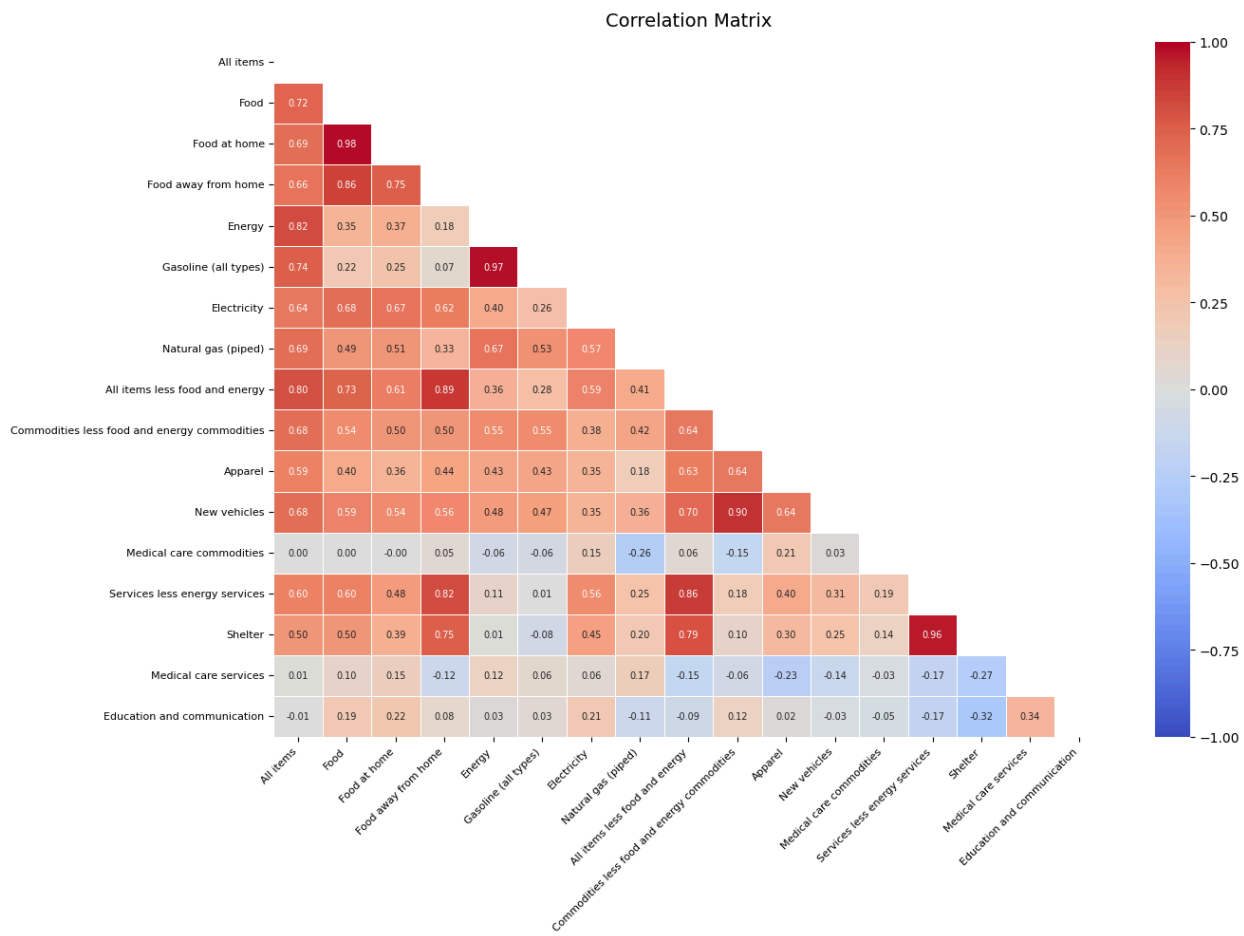
- **Source & Processing:**
  Obtained from BLS CPI dataset. Standardized during preprocessing.
- **Observations: ~554**
- **Data Type: Quantitative**
- **Role: Independent variable**

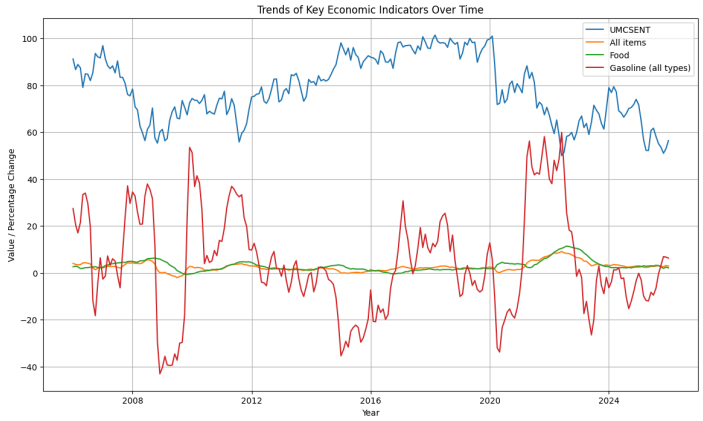
- **Notes:**
    - **Captures healthcare service costs**

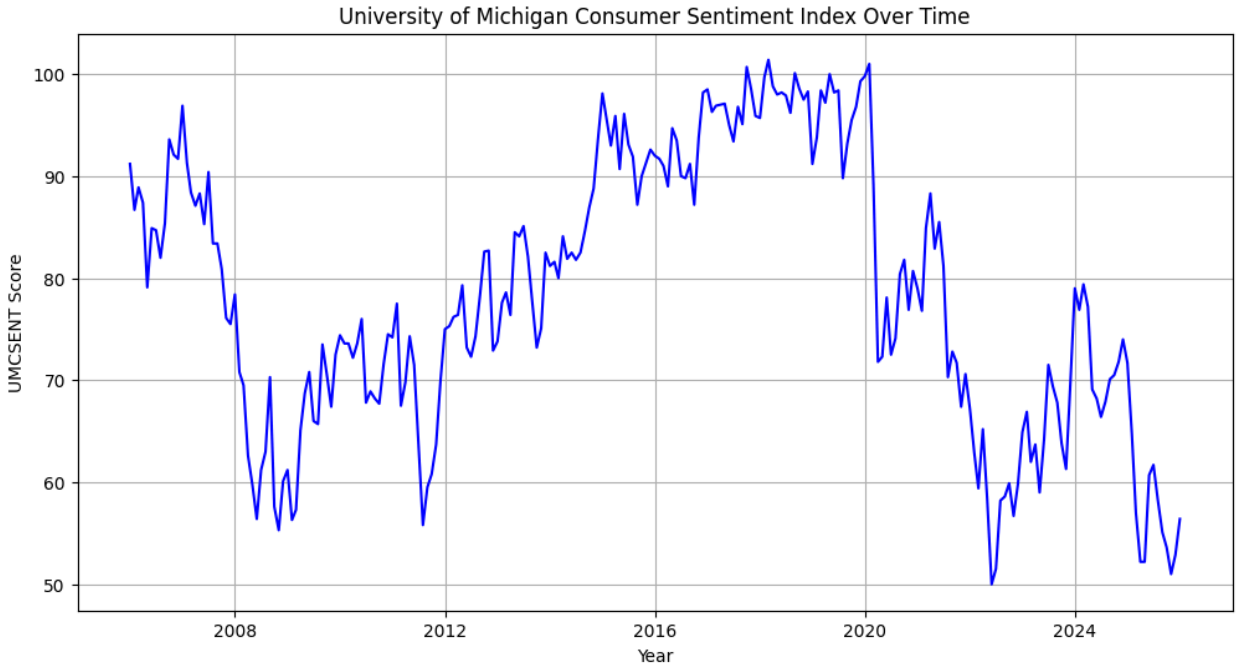
Data Cleaning & Preprocessing Summary

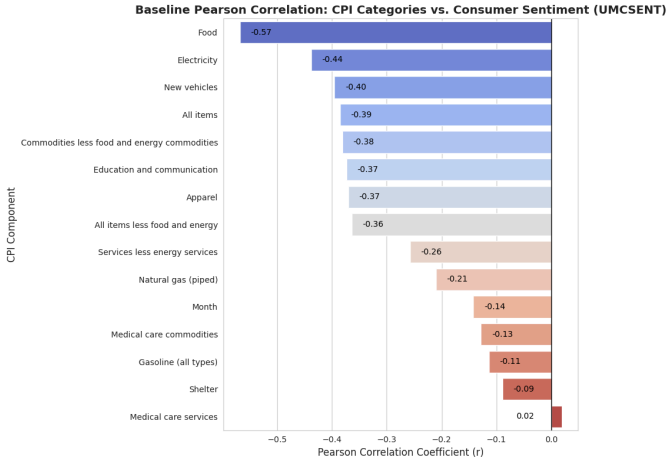
- **Merged CPI and UMCSENT datasets using monthly timestamps**
- **Removed missing values (e.g., October 2025)**
- **Standardized CPI features**
- **Removed multicollinear variables using correlation threshold ($|r| \geq 0.85$)**
- **Identified a structural break around 2020 (COVID impact)**

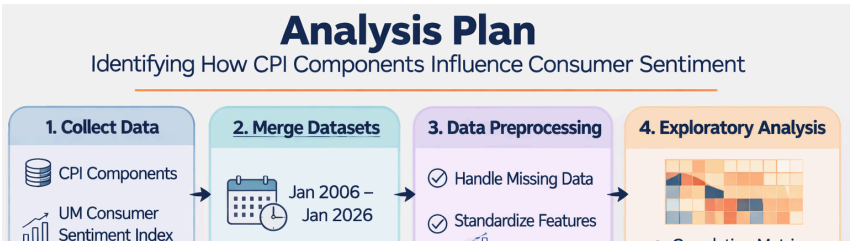
Visualization:




Trends of Key Economic Indicators Over Time
UMCSENT
All items
Food
Gasoline (all types)
Value / Percentage Change
Year
100
80
60
40
20
0
−20
−40
2008
2012
2016
2020
2024


University of Michigan Consumer Sentiment Index Over Time
UMCSENT Score
Year
100
90
80
70
60
50
2008
2012
2016
2020
2024


Baseline Pearson Correlation: CPI Categories vs. Consumer Sentiment (UMCSENT)
CPI Component
Pearson Correlation Coefficient (r)
Food -0.57
Electricity -0.44
New vehicles -0.40
All items -0.39
Commodities less food and energy commodities -0.38
Education and communication -0.37
Apparel -0.37
All items less food and energy -0.36
Services less energy services -0.26
Natural gas (piped) -0.21
Month -0.14
Medical care commodities -0.13
Gasoline (all types) -0.11
Shelter -0.09
Medical care services 0.02
−0.5
−0.4
−0.3
−0.2
−0.1
0.0


Analysis Plan
Identifying How CPI Components Influence Consumer Sentiment
1. Collect Data
CPI Components
UM Consumer Sentiment Index
2. Merge Datasets
Jan 2006 – Jan 2026
3. Data Preprocessing
Handle Missing Data
Standardize Features
4. Exploratory Analysis